Time & Work,
Pipes & Cisterns

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Time & Work, Pipes & Cisterns

- ✓ Basic Concept
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BASIC CONCEPT

More men can do more work.

More work means more time required to do work.

More men can do more work in less time.

M men can do a piece of work in T hours, then Total effort or work = MT man hours.

If A can do a piece of work in D days, then A's 1 day's work = 1/D.

Part of work done by A for t days = t/D.

If A's 1 day's work = 1/D, then A can finish the work in D days.

MDH/W = Constant

where, M = Number of men, D = Number of days

H = Number of hours per day,

W = Amount of work

BASIC CONCEPT

If M1 men can do W1 work in D1 days working H_1 hours per day and M_2 men can do W_2 work in D_2 days working H_2 hours per day,

then $(M_1 \times D_1 \times H_1)/W_1 = (M_2 \times D_2 \times H_2)/W_2$

TYPES OF QUESTIONS

Type 1:
A takes 10 days to complete a piece of work. B takes 15 days to complete the same piece of work. How much time will A and B together take to complete the same work?

One Day/Unitary Method

A takes 10 days to complete the work, thus in 1 day A can do 1/10 units of work.

B takes 15 days to complete the work, thus in 1 day B can do 1/15 units of work.

Together, A & B completes
1/10 + 1/15 units of work in 1 day
= 1/10 + 1/15 =

(10 + 15)/(10x15)
= 25/150 = 1/6

Therefore, to complete the work, they will need 6 days.

TIME & WORK

1. A, B and C alone can complete a work in 10, 20 and 30 days respectively. How much time they will take if they are doing the work together.

   A. 60/7 B. 60/11 C. 120/11 D. 120/7

	A	B	C
No of days required to complete the work	10	20	30
Work done in 1 day	1/10	1/20	1/30

Total Work done by A, B and C in 1 day

$$\frac{1}{10} + \frac{1}{20} + \frac{1}{30}$$

$$= \frac{6+3+2}{60}$$

$$= \frac{11}{60}$$

Hence, no of days required

$$= \frac{60}{11} \text{ days}$$

$$= 5\frac{5}{11} \text{ days}$$

2. A, B and C alone can complete a work in 10, 15 and 20 days respectively. How much time they will take if they are doing the work together.

A. 60/11 B. 60/17 C. 60/18 D. 60/13

	A	B	C
No of days required to complete the work	10	15	20
Work done in 1 day	1/10	1/15	1/20

Total Work done by A, B and C in 1 day $\frac{1}{10} + \frac{1}{15} + \frac{1}{20}$

$$= \frac{6+4+3}{60}$$

$$= \frac{13}{60}$$

Hence, no of days required

$$= \frac{60}{13} \text{ days}$$

$$= 4\frac{8}{11} \text{ days}$$

TYPES OF QUESTIONS

Type 2:
A takes 10 days to complete a piece of work.
A and B together completes the same work in 6 days.
How much time will B take to complete the same work?

One Day/Unitary Method

A takes 10 days to complete the work, thus in 1 day A can do 1/10 units of work.

A & B takes 6 days to complete the work, thus in 1 day they can do 1/6 units of work.

Let the time taken by B is x days. In 1 day,
$1/10 + 1/x = 1/6$

Solving, we get $x = 15$ days.

Therefore, to complete the work, B will take 15 days.

4. A and B can complete a work in 20 days , B and C in 30 days and C and A in 40 days. In how many days can they complete the same work, if they work together?

A. 240/13 days B. 220/13 days C. 260/11 days D. 220/11 days

A and B can complete a work	20 days	So, work done by A and B in 1 day	1/ 20
B and C can complete a work	30 days	So, work done by B and C in 1 day	1/ 30
A and C can complete a work	40 days	So, work done by A and C in 1 day	1/ 40

So, work done by 2A, 2B and 2C in 1 day $\frac{1}{20}+\frac{1}{30}+\frac{1}{40} = \frac{13}{120}$

So, work done by A, B and C in 1 day $= \frac{13}{120} \times \frac{1}{2} = \frac{13}{240}$

No of days required by A, B and C $= \frac{240}{13} = 18\frac{6}{13}$ days

TYPES OF QUESTIONS

Type 3:
A takes 10 days to complete a piece of work and B takes 20 days to complete the same work. A and B together works for 2 days and then A leaves and the rest of the work has to be completed by B alone. How much time will B take to complete the remaining work?

One Day/Unitary Method

A takes 10 days to complete the work, thus in 2 day **A** can do 2/10 units of work.

B takes 20 days to complete the work, thus in 2 day B can do 2/20 units of work.

Let the time taken by **B** to compete the remaining work is x days.

$2/10 + 2/20 + x/20 = 1$
Solving, we get $x = 14$ days.

For the remaining work, B has to work alone. Therefore, to complete the work, B will take = 14 days

3. A and B can complete a work in 20 & 30 days respectively. They start to work together, but after 2 days, A left the work. In how man days remaining work will be done by B?

A. 20 days B. 22 days C. 24 days D. 25 days

5. A, B and C alone can complete a work in 12, 16 and 20 days respectively. They start the work together, but after 4 days A and B leave the work. In how many days C can complete the remaining work?

A. 13/2 days B. 17/2 days C. 13/3 days D. 17/3 days

Let C alone can complete the remaining work in x days.

then ATQ

$$\frac{4}{12}+\frac{4}{16}+\frac{4}{20}+\frac{x}{20}=1$$

$$\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{x}{20}=1$$

$$\frac{20+15+12+3x}{60}=1$$

$$47+3x=60$$

$$x=\frac{13}{3}\text{ days}$$

6. A and B can complete a work in 15 & 20 days respectively. Both start the work but B leaves it 5 days before the Completion of the work. For how many days did they work together?

A. 45/7 days B. 40/7 days C. 35/8 days D. 30/8 days

Let they work together for x days.

So ATQ
$$\frac{x}{15}+\frac{x}{20}+\frac{5}{15}=1$$
$$\frac{4x+3x+20}{60}=1$$
$$4x+3x+20=60$$
$$7x=40$$
$$x=\frac{40}{7}\text{days}$$
$$x=5\frac{5}{7}\text{days}$$

7. A can do a work in 20 days while B in 30 days. They work on alternate days, with A starting the work. In how many days can the work be done?

A. 20 days **B. 22 days** **C. 24 days** **D. 30 days**

They are working alternately

So, work done by A and B in 2 day $\mathbf{\frac{1}{20} + \frac{1}{30} = \frac{1}{12}}$

So, work done by A and B in 1 day $\mathbf{= \frac{1}{24}}$

So, number of days required to complete the work $\boldsymbol{= 24\ days}$

TYPES OF QUESTIONS

Type 4:
4 men work for 5 hours a day to finish 2 units of work, in 24 days. How many days will 6 men need to finish 3 units of work working 6 hours a day?

Things to Remember:

Men $\propto$ Work
Hrs/day $\propto$ Work
Days $\propto$ Work

Days 1/$\propto$ Hrs/day

Men	Work	Hrs/day	Days
4	2	5	24
6	3	6	d

d = 24 x 4/6 x 3/2 x 5/6

Hence,
d = 20 days.

8. The ratio of work capacities of A and B is 2:1. What will be the ratio of the time taken by them in completing the same work?

A. 4:1 B. 1:4 C. 1:2 D. 2:1

the ratio of the time taken by them in completing the same work **1:2**

Days $1/\propto$ Hrs/day

9. If 30 men can complete a work in 25 days, then in how many days, will 75 men complete the same work?

A. 10 days **B. 15 days** **C. 20 days** **D. 30 days**

$$Days = 25 \times \frac{30}{75}$$

$$= 10\ Days$$

10. 40 men complete a work in 60 days. 20 men leave the work after 30 days. In how many days will the remaining work be completed?

A. 40 days **B. 60 days** **C. 80 days** **D. 100 days**

Men	**Work**	**Days**
40	1	60
20	1/2	d

$$d = 60 \times \frac{40}{20} \times \frac{1}{2}$$

d=60 days

11. A can complete a piece of work in 12 days and B can complete the same piece of work in 16 days. In how many days can A and B together complete the same piece of work?

A. 40/7 days B. 44/5 days C. 46/5 days D. 48/7 days

12. 8 males can complete a work in 12 days and 6 females take 32 days to complete the same work. If 8 males and 8 females work together, then in how many days the work will be done?

A. 7 days B. 8 days C. 9 days D. 10 days

8 males can complete a work =12 days

So work done by 8 males in 1 day =1/ 12

So work done by 1 males in 1 day $= \dfrac{1}{12 \times 8}$

Now ATQ, work done by 8 males in 1 day $= \dfrac{8}{12 \times 8} = \dfrac{1}{12}$

Similarly, work done by 8 females $= \dfrac{8}{32 \times 6} = \dfrac{1}{24}$

So, work done by 8 males and 8 females in 1 day $= \dfrac{1}{12} + \dfrac{1}{24} = \dfrac{3}{24}$

So, number of days required $= \dfrac{24}{3} = 8\ days$

13. 15 men complete a work in 24 days while 12 women can complete it in 36 days. In how many days 12 Men with 18 Women can complete this work.

A. 36/5 days **B. 38/5 days** **C. 40/3 days** **D. 48/3 days**

ATQ, work done by 12 men & 18 women in 1 day $= \frac{1 \times 12}{24 \times 15} + \frac{1 \times 18}{36 \times 12}$

$$= \frac{1}{30} + \frac{1}{24}$$

$$= \frac{9}{120} = \frac{3}{40}$$

So no of days required by 12 men & 18 women to complete the work $= \frac{40}{3}\, days$

14. 20 Examiners check 600 answer copies in 25 days. Then in how many days 30 examiners can check 400 answer Copies.

A. 50/3days **B. 100/9 days** **C. 110/9 days** **D. 80/3 days**

Examiners	copies	days
20	600	25
30	400	x

$$x = \frac{20}{30} \times \frac{400}{600} \times 25$$

$$x = \frac{100}{9}\text{days}$$

15. 26 men complete a piece of work in 17 days. How many more men must be hired to complete the same work in 13 days

A. 6 **B. 10** **C. 8** **D. 12**

Number of men required to complete the work in 13 days $= 26 \times \frac{17}{13}$

$= 34\ mens$

So, Number of more men required to complete the same work in 13 days $= 34\ mens \ -26\ mens$

$= 8\ mens$

TYPES OF QUESTIONS

Type 5: (In terms of Efficiency)
If the efficiency relation for A, B & C is A= 2B = 3C And A, B & C can finish a piece of work in 24 days.

Since the efficiency term is A=2B=3C,

means if B takes B days to finish the task, A & C will take B/2, 3B/2 days respectively.

Thus, the equation will be,

$$1/A + 1/B + 1/C = 1/24$$

$$2/B + 1/B + 2/3B = 1/24$$

Solving the equation, we get B= 88 Days

16. A is thrice as fast as B. Together they can do a job in 15 days. In how many days will B finish the work?

A. 45 days B. 60 days C. 30 days D. 50 days

17. P and Q can do a work in 20 and 30 days respectively. While doing the work together they get an amount of Rs. 50,000. What is the share of P in this amount?

A. Rs. 20,000 B. Rs. 25,000 C. Rs. 30,000 D. Rs. 40,000

18. A man, a woman and a boy together complete a piece of work in 5 days. If a man alone can do it in 12 days and a woman alone in 15 days. How long will a boy take to complete the work?

A. 15 days **B. 20 days** **C. 25 days** **D. 30 days**

19. If 16 workers working 6 hours a day can build a wall of Length 150 m, Breadth 20 m and height 12 m in 25 days. In how many days 12 workers working 8 hours a day can build a wall of Length 800 m, Breadth 15 m and Height 6 m?

A. 20 days B. 30 days C. 40 days D. 50 days

20. Mr. Modi can copy 40 Pages in 10 minutes. Mr. Xerox and Mr. Modi, both working together can copy 250 pages in 25 minutes. So, in how many minutes Mr. Xerox can copy 36 pages.

A. 5 mins **B. 6 mins** **C. 8 mins** **D. 4 mins**

21. Mohan can do a work in 15 days. After working for 3 days he is joined by Vinod. If they complete the remaining work in 3 more days, in how many days can Vinod alone complete the work?

A. 10 days B. 8 days C. 5 days D. 12 days

22. Arun can do a certain work in the same time in which Bipasha and Rahul together can do it. If Arun and Bipasha together could do it in 10 days and Rahul alone in 50 days, then Bipasha alone could do it in:

A. 15 days B. 20 days C. 25 days D. 30 days

23. Sekar, Pradeep and Sandeep can do a piece of work in 15 days. After all the three worked for 2 days, sekar left. Pradeep and Sandeep worked for 10 more days and Pradeep left. Sandeep worked for another 40 days and completed the work. In how many days can sekar alone complete the work if sandeep can complete it in 75 days?

A. 25 days B. 20 days C. 30 days D. 35 days

24. Dinesh does 80% of a work in 20 days. He then calls in Gokul and they together finish the remaining work in 3 days. How long Gokul alone would take to do the whole work?

A. 39 days B. 37 days C. 37.5 days D. 40 days

25. Hari and Vijay can together finish a work in 30 days. They worked together for 20 days and then Vijay left. After another 20 days, hari finished the remaining work. In how many days hari alone can finish the work?

A.45 B.60 C.35 D.50

26. A work could be completed in 100 days. However due to absence of 10 workers, it was completed in 110 days. The original number of workers was

A. 110 B. 110 C. 55 D. 50

27. A can do a piece of work in 10 days. A undertook to do it for Rs. 320. With the help of B he finishes the work in 8 days. The B's share is?

A. Rs. 80 B. Rs. 64 C. Rs. 100 D. Rs. 120

28. If the wages of 6 men for 15 days be Rs. 1400, then the wages for 9 men for 12 days will be

A. Rs. 1680 B. Rs. 1840 C. Rs. 1050 D. Rs. 1600

29. Amar can lay a railway track between two given stations in 32 days and Manoj can do the same job in 24 days. With the help of Akash, they did the job in 8 days only. Then Akash alone can do the job in?

A. 96/5 days B. 20 days C. 48/5 days D. 10 days

30. Virat can do a piece of work in 40 days. Rohit is 25% more efficient than Virat. The number of days taken by Rohit to do same piece of work is?

A. 30 days **B. 32 days** **C. 35 days** **D. 36 days**

PIPE & CISTERN

All the time and work concepts are as it is followed in Pipes and Cisterns.

The work done in filling a cistern is taken as positive and the work done in emptying a cistern is taken as negative.

Example Taps X and Y can fill a tank in 30 and 40 minutes respectively. Tap Z can empty the filled tank in 60 minutes. If all the three taps are kept open for one minute each, how much time will the taps take to fill the tank?

X's one minute work = 1/30

Y's one minute work = 1/40

Z's one minute work = -1/60

They work for consecutive minutes, in first 3 minutes they finish,

1/30 + 1/40 – 1/60 = 1/24,

means in 3 minutes they finish 1/24 work

Thus, to finish the entire work, they will need 70 1/3 minutes.

31. Two pipes P and Q, will fill a cistern in 16 and 20 minutes respectively. If both pipes are opened, find when the first pipe must be turned off so that the cistern may be just filled in 10 minutes

A. After 17/2 min **B. After 10 min**
C. After 12 min **D. After 8 min**

32. A pipe can fill a tank within 2 hours. Because of a leak it took 7/3 hours to fill the tank. The leak can drain the entire tank in

A. 14 hours B. 12 hours C. 16 hours D. 10 hours

33. A cistern is filled in 6 hours and it takes 8 hours when there is a leak in the bottom. If the cistern is full, in what times shall the leak empty it.

A. 48 hours B. 36 hours C. 24 hours D. 12 hours

34. A cistern is normally filled in 8 hours but it takes two hours longer to fill because of a leak in its bottom. If the cistern is full, the leak will empty it in

A. 16 hours B. 40 hours C. 20 hours D. 25 hours

35. Two pipes A and B can fill a tank in 24 min and 32 min respectively. If both the pipes are opened simultaneously after how much time B should be closed so that the tank is full in 18 min.

A. 6 min **B. 8 min** **C. 10 min** **D. 12 min**

36. A tank is filled in 20 hours by three pipes A, Band C. The pipe C is twice as fast as B and B is twice as fast as How much time will pipe A alone take to fill the tank?

A. 100 hours B. 120 hours C. 140 hours D. 200 hours

37. A tap can fill a tank in 12 hours. After half the tank is filled then 3 more similar taps are opened. What will be the total time taken to fill remaining half of the tank?

A. 1 hour B. 1 hour 15 min C. 1 hour 30 min D. 1 hour 45 min

38. Bucket A has thrice the capacity as bucket B. It takes 90 turns for bucket A to fill the empty drum. How many turns it will take for both the buckets A and B, with each turn together to fill the empty drum?

A. 60 turns B. 64 turns C. 66 turns D. 68 turns

39. Taps P, Q, and R are connected to a water tank and the rates of flow of water are 42 liters per hour, 56 liters per hour and 48 liters per hour respectively. Taps P and Q fill the tank while tap R empties the tank. If all the three taps are opened simultaneously, the tank gets filled up in 16 hours, then what is the capacity of the tank?

A. 900 liters B. 800 liters C. 1000 liters D. 816 liters

40. A tank is fitted with 4 pipes. Out of 4 pipes, 3 are inlet pipes and 1 is outlet pipe. One inlet pipe will take 4 hours to completely fill the tank. Similarly one outlet pipe will take 4 hours to empty the tank. If the tank is empty initially, how much time will it take to completely fill the tank with all pipes turned on?

A. 4 hours B. 3 hours C. 2 hours D. 1 hours